FIITJEE

ALL INDIA TEST SERIES

PART TEST – I

JEE (Main)-2019-20

TEST DATE: 9-11-2019

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A

(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. For a particle projected with initial velocity $\vec{u} = 6\hat{i} + 8\hat{j} + 20\hat{k}$, find angular velocity of the particle about the point of projection, at t = 2 sec. (Given $\vec{g} = -10\hat{k}$)

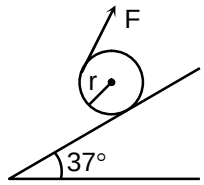
   (A) $\frac{1}{5}\hat{i} - \frac{3}{20}\hat{j}$

   (B) $-\frac{1}{5}\hat{i} + \frac{3}{20}\hat{j}$

   (C) $\frac{\sqrt{2}}{5}\hat{i} - \frac{3\sqrt{2}}{20}\hat{j}$

   (D) $\frac{\sqrt{2}}{5}\hat{i} + \frac{3\sqrt{2}}{20}\hat{j}$

2. A solid cylinder of mass 'm' is kept stationary on a fixed incline of 37°, by applying a force tangentially as shown in the figure. Calculate the minimum coefficient of friction that the surface must have in order to achieve equilibrium of the cylinder by applying a force of minimum magnitude.



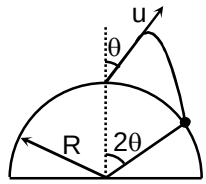
   (A) $\frac{3}{4}$

   (B) $\frac{3}{8}$

   (C) $\frac{3}{16}$

   (D) none of these

3. For what value of initial speed u, a projectile launched at θ (from the vertical) from the top point of a hemisphere will land at 2θ angular position(from the vertical) on the hemisphere as shown in the figure.



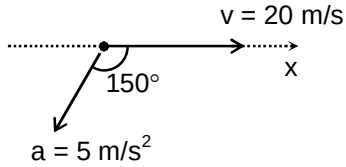
   (A) $\sqrt{gR}\cot\theta$

   (B) $\sqrt{gR}\frac{\cos^2\theta}{\sin\theta}$

   (C) $\sqrt{gR}\cos\theta$

   (D) Not possible

4. A particle moving in space with a velocity 20 m/s along positive x-axis starts experiencing a constant acceleration at t = 0 as shown in the figure. The minimum value of the ratio $\frac{R_{min}}{R_{max}}$, where R_{min}/R_{max} is the minimum and maximum radius of curvature of the particle's trajectory for any t > 0 is



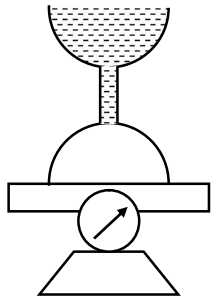
(A) $\frac{1}{8}$

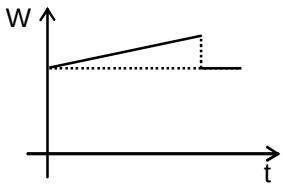
(B) $\frac{1}{6}$

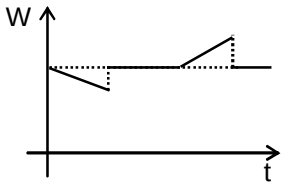
(C) $\frac{1}{5}$

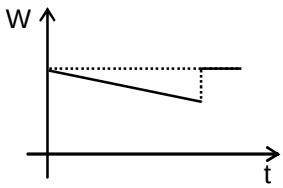
(D) none of these

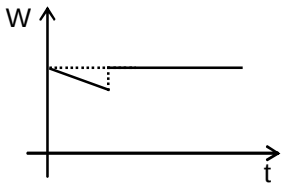
5. An hour glass containing sand is kept on a weighing scale. At t = 0, sand starts falling from upper compartment to lower compartment at constant rate. How will the reading of scale look like when plotted against time.



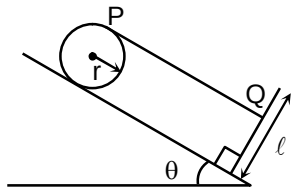
(A) 

(B) 

(C) 

(D) 

6. A rod of length 'ℓ' and cylinder of radius 'r' is kept on an incline plane as shown in the figure. The rod is pivoted, while the cylinder can roll without slipping. A light string PQ attaches top of cylinder to some point on the rod such that it is parallel to the incline. The minimum value of radius of cylinder to ensure that the string is taut when the system is released from rest is



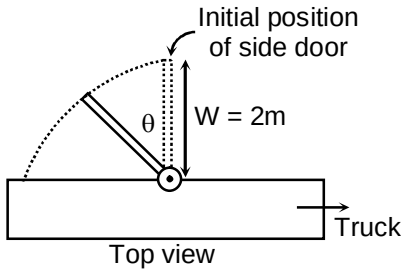
(A) $\frac{2}{5}\ell$

(B) $\frac{21}{50}\ell$

(C) $\frac{4}{9}\ell$

(D) none of these

7. A truck has its side door initially open as shown. The dimension of the door is 2m $\times$ 3m (W $\times$ ℓ) and its mass is M = 5 kg. At t = 0, the truck starts moving with uniform acceleration a = 5 m/sec^2 as shown in figure (top view). At any time t, if the door makes angle θ with its initial position, then the component of force exerted by hinge on the door along its width is



(A) $\frac{125}{2}\sin\theta$

(B) 250 sin θ

(C) 75 sin θ

(D) none of these

8. For a particle moving in space with velocity $\vec{v}$, which of the following is incorrect.

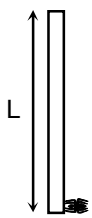
(A) $\left|\frac{d\vec{v}}{dt}\right| \geq \frac{d|\vec{v}|}{dt}$

(B) $\int_0^t |\vec{v}|dt \geq \left|\int_0^t \vec{v}\cdot dt\right|$

(C) $|\Delta\vec{v}| \leq \int_{t_1}^{t_2} |\vec{a}|\,dt$

(D) $\frac{d|\vec{v}|}{dt} = \frac{d\vec{v}}{dt}\cdot\left(\frac{\vec{v}}{v}\right)$

9. A stick of length L is dropped from rest. At same instant, an insect starts moving up the stick with a constant speed u relative to the stick. The maximum height attained by the insect from its initial position with respect to the ground is



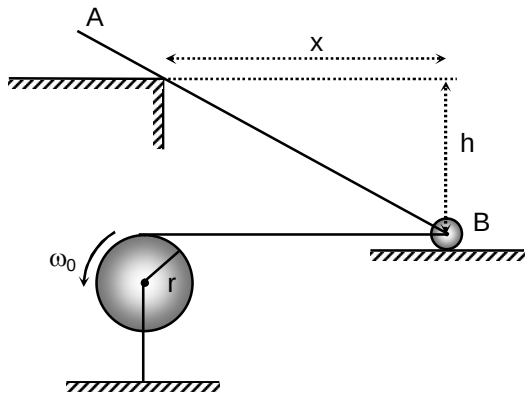
(A) $\frac{u^2}{2g}$

(B) $\frac{u^2}{g}$

(C) $\frac{2u^2}{g}$

(D) none

10. In the arrangement shown, a bar AB is connected through a rotating drum via a massless string which is attached to the end B of the bar. The portion of the rod near end A rests on a horizontal surface. The drum rotates with a uniform angular speed ω_0. The angular speed of bar AB at given position is



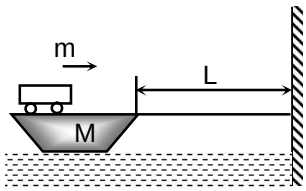
(A) $\frac{\omega_0 rx}{\sqrt{x^2+h^2}}$

(B) $\frac{\omega_0 rh}{(x^2+h^2)}$

(C) $\frac{\omega_0 rh}{\sqrt{x^2+h^2}}$

(D) $\frac{\omega_0 rx}{(x^2+h^2)}$

11. A car of mass m is initially at rest on a boat of mass M and is tied to the shore as shown. The car starts accelerating from rest at t = 0 and acquires a velocity v_0 in time t_0. At $t = t_0$, the car applies brake and comes to rest relative to the boat in no time. Neglect friction between the boat and the water. The time 't' in which the boat strikes the wall is



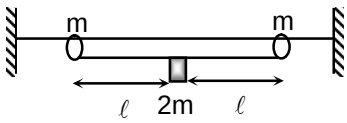
(A) $t_0 + \frac{1}{v_0}$

(B) $t_0 + \frac{L(M+m)}{mv_0}$

(C) $\frac{LM}{mv_0}$

(D) none of these

12. Two small rings of mass 'm' each are contained to move on smooth wire. The rings are connected to a block of mass '2m' via massless strings of length 'ℓ' each. Initially, the system is held at rest as shown. The velocity of block when string makes 60° with the vertical is



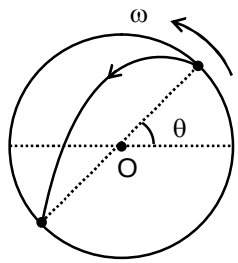
(A) $\sqrt{2g\ell}$

(B) $\sqrt{\frac{3g\ell}{5}}$

(C) $\sqrt{\frac{g\ell}{5}}$

(D) $\sqrt{\frac{3g\ell}{4}}$

13. A spinning drum of radius 'R' with its axis parallel to the ground, carries dust particles deposited on its inner surface. Find minimum angular speed with which drum be rotated so that a dust particle separating from the drum may fall at its diametrical opposite position in space. Assume, sufficient friction present to prevent any relative slipping of dust particle on the drum.



(A) $\sqrt{\frac{g}{2R}}$

(B) $\sqrt{\frac{g}{\sqrt{2}R}}$

(C) $\sqrt{\frac{g}{4R}}$

(D) None of these

14. A particle moves in xy plane. The position vector of particle at any time t is $\vec{r} = \left\{(2t)\hat{i} + (2t^2)\hat{j}\right\}$ m.

The rate of change of θ at t = 2 sec, where θ is the angle made by the velocity vector from the positive x-axis, is

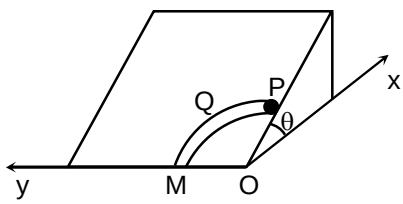
(A) $\frac{1}{14}$ rad/sec

(B) $\frac{2}{17}$ rad/sec

(C) $\frac{4}{7}$ rad/sec

(D) $\frac{6}{5}$ rad/sec

15. A circular groove PQM of radius R is cut in an inclined plane as shown. A small ball of mass m is released form P. The magnitude of the displacement of wedge by the time ball reaches M is ($\theta = 60°$, R = 1m, m = 2kg, M = 3kg)



(A) $\sqrt{5}$ m

(B) $\sqrt{\frac{1}{5}}$ m

(C) $\sqrt{\frac{17}{5}}$ m

(D) none of these

16. A man is rotating a stone of mass 10 kg tied at the end of a light rope in a circle of radius 1m. To do this, he continuously moves his hand in a circle of radius 0.6 m. Assume, both circular motions to be occurring in the same horizontal plane. What is the maximum speed with which he can throw the stone, if he can exert a pull not exceeding 1250 N on the string.

(A) $10\sqrt{2}$ m/s

(B) $5\sqrt{5}$ m/s

(C) 10 m/s

(D) 20 m/s

17. A man of mass m while jumping up in the air applies a force $F = F_0 \cos \omega t$ on the ground. Assume, that he remains in contact with the ground from t = 0 to $t = \frac{\pi}{2\omega}$ sec.

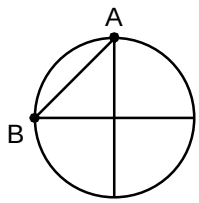
(A) Velocity of man during take off $\left(t = \frac{\pi}{2\omega}\right)$ is less than $\frac{F_0}{m\omega}$

(B) Man is producing peak power at the instant of take off.

(C) The average power produced by man is zero.

(D) The ratio of power produced at $t = \frac{\pi}{8\omega}$ and $t = \frac{3\pi}{8\omega}$ is more than 1.

18. Two beads A and B of equal mass 'm' are connected by a light string. The beads are positioned to move on a smooth ring in a vertical plane as shown in the figure. The tension in the string just after release is



(A) $\sqrt{2}$ mg

(B) $\frac{mg}{2}$

(C) $\frac{mg}{4}$

(D) $\frac{mg}{\sqrt{2}}$

19. A particle A is fixed at the origin of a fixed coordinate system. B is another particle that experiences a force $\vec{F} = \left(-\frac{2\alpha}{r^3} + \frac{\beta}{r^2}\right)\hat{r}$ due to particle A, where $\vec{r}$ is the position vector of B with respect to A. Find the work done in moving particle B slowly from $P_1\left(\sqrt{2}r_0, \sqrt{2}r_0\right)$ to point $P_2\left(\frac{r_0}{\sqrt{2}}, \frac{r_0}{\sqrt{2}}\right)$ by an external agent, where r_0 is the equilibrium position of the particle.

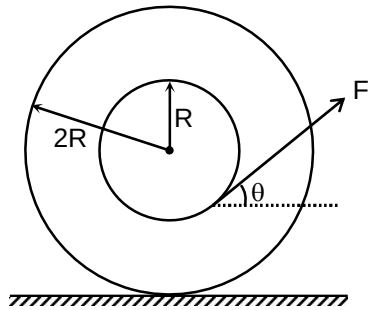
(A) $\frac{9\beta^2}{64\alpha}$

(B) $\frac{\beta^2}{16\alpha}$

(C) $-\frac{\beta^2}{16\alpha}$

(D) $-\frac{9\beta^2}{64\alpha}$

20. A spool of mass 'm' has inner and outer radius R and 2R respectively. A thread is wound on the inner disc and its free end is pulled with a force F at an angle θ with the horizontal in the direction as shown in the figure. Assume no slipping condition for thread and the spool and $I_{cm} = 2mR^2$, then



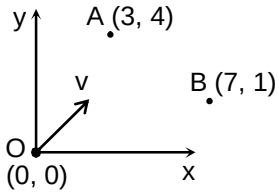
(A) For $0 \le \theta \le 90°$, frictional force acts leftward only.
(B) For $0 \le \theta \le 60°$, it acts leftward and for $60° \le \theta \le 90°$ it acts rightward.
(C) For a certain θ between 0° and 90° friction becomes zero.
(D) none of these

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. Two men A and B of mass M and 2M are standing 120° apart on a circular platform of mass 4M and radius R. The platform is free to rotate about its centre, while men are standing at periphery. Now men interchange their position with respect to disc by walking along periphery only. If $\Delta\theta$ represents magnitude of angle rotated by platform with respect to the ground, find the ratio $\frac{\Delta\theta_{max}}{\Delta\theta min}$.

22. A(3 m, 4 m) and B(7 m, 1 m) are two coins on a carrom board. The striker is placed at origin O. If the striker can be shot with a speed of 5 m/sec, then find the minimum time (in sec) taken by striker to become collinear with coin A and B.

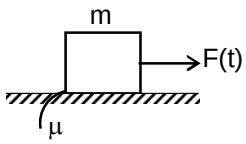


SECTION – C
(Numerical Answer Type)

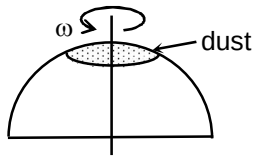
This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. The resultant of two vectors $\vec{A} = 5\hat{i}$ and $\vec{B}$ is $\vec{R}$. The direction of $\vec{R}$ is such that it makes an angle $\frac{\pi}{6}$ with the positive x-axis in anticlockwise sense. Find the magnitude of $\vec{R}$ when $\vec{B}$ has minimum magnitude. (Take $\sqrt{3} = 1.732$)

24. A block of mass m = 4 kg is kept on a rough surface of $\mu = 0.25$. At t = 0, an external force F(t) that varies linearly with time starts acting on the block. The force acts for a duration of 8 sec only and F(t = 8) = 0 N. If the block starts moving at t = 2 sec and finally stops at t = 11 sec, then find the time (in sec) instant when velocity acquired by the block is maximum.



25. A hemisphere of radius R = 0.5 m is rotated with a uniform angular speed $\omega = \sqrt{10}$ rad/sec for some time. The hemisphere is then stopped and it was found that only 20% of its top surface area remained covered with dust particles. Find the coefficient of friction between the particles and the hemisphere.



Chemistry

PART – II

SECTION – A

(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. The geometry around each iodine in dinuclear anion $\left[I_2(OH)_2O_8\right]^{4-}$ is:
    (A) Octahedral
    (B) Monocapped octahedral
    (C) Square pyramidal
    (D) Pentagonal bipyramidal

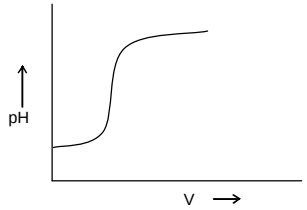
27. Amongst Zn, Ga, Ge and As the element with the lowest first I.E. is:
    (A) As
    (B) Zn
    (C) Ga
    (D) Ge

28. The magnetic property of which molecule changes if we assume s – p mixing is not operational for all given molecules:
    (A) N_2
    (B) F_2
    (C) O_2
    (D) C_2

29. Consider the following reversible reaction; $A(g) + 2B(g) \rightleftharpoons AB_2(g)$ $\quad K_c = \frac{1}{2}$.

    The above equilibrium is established in a 1 L flask and at equilibrium 2 moles of each A and B are present. If 2.0 moles of B are added further how many moles of AB_2 should be added so that moles of A does not change?
    (A) 6
    (B) 8
    (C) 10
    (D) 12

30. During titration of acetic acid with aq. NaOH solution, the neutralization graph has a vertical line. This line indicates:



    (A) Neutral nature at equivalence point
    (B) Acidic nature at equivalence point
    (C) Depends on experimental proceedings
    (D) Alkaline nature at equivalence point

31. Which is the poorest reducing agent?
(A) Nascent hydrogen
(B) Atomic hydrogen
(C) Dihydrogen
(D) All have same reducing strength

32. For which of the following reactions, the degree of dissociation cannot be calculated from the V.D. data:

I. $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

II. $2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

III. $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$

IV. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

(A) I and III
(B) II and IV
(C) I and II
(D) III and IV

33. On dissolving moderate amount of sodium metal in liquid NH_3 at low temperature, which of the following does not occur?
(A) Blue coloured solution is obtained.
(B) Na^+ ions reformed in the solution.
(C) Liquid ammonia solution becomes good conductor of electricity.
(D) Liquid ammonia solution becomes diamagnetic.

34. Consider the following statements about carbon suboxide
I. It is a linear molecule.

II. It is an optically active cumulene.

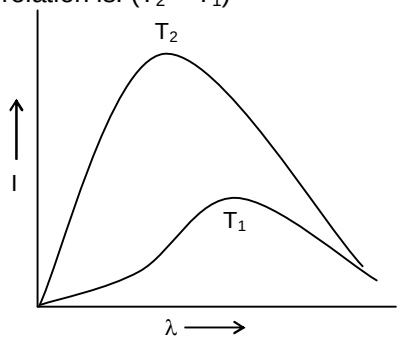
III. The total number of σ bonds, π bonds and lone pair in the compound are 12.
The correct statement is/are:
(A) I and II
(B) II and III
(C) I and III
(D) All of the above

35. In the compound M – O – H, the M – O bond will be broken if:
(A) ΔE.N. of M and O < ΔE.N. of O and H
(B) ΔE.N. of M and O = ΔE.N. of O and H
(C) ΔE.N. of M and O > ΔE.N. of O and H
(D) Cannot be predicted by ΔE.N. data

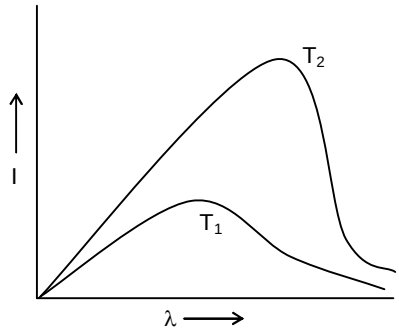
36. Choose the INCORRECT statement:
(A) Amorphous solids are short range orders
(B) Amorphous solids are anisotropic in nature
(C) Crystalline solids have definite heat of fusion
(D) Crystalline solids are true solids

37. The correct observed curve for a backbody at different temperature representing intensity wavelength relation is: ($T_2 > T_1$)

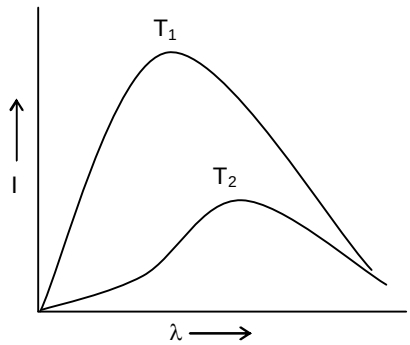
(A)



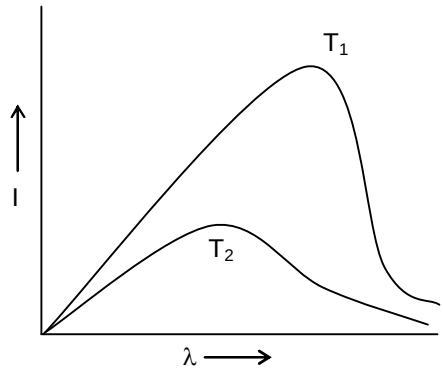
(B)



(C)



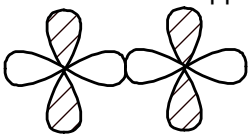
(D)



38. The oxidation state of sulphur in the product/(s) of the reaction of S_8 in basic medium is/are:

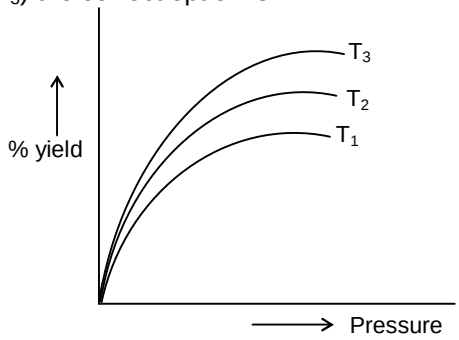
(A) +2 and +6
(B) - 2 and +6
(C) - 2 and +2
(D) - 2 only.

39. The orbital overlapping in the figure results in:



(A) d – d π antibonding
(B) d – d σ bonding
(C) d – d σ antibonding
(D) d – d π bonding

40. The preparation of $SO_3(g)$ by reaction $SO_2(g)+\frac{1}{2}O_2(g)\rightleftharpoons SO_3(g)$ is an exothermic reaction following temperature – pressure relationship for its % yield, for temperatures (T_1, T_2 and T_3) the correct option is:



(A) $T_3 > T_2 > T_1$
(B) $T_1 = T_2 = T_3$
(C) $T_1 > T_2 > T_3$
(D) Not enough information.

41. The existence of a solid compound in more than one modification is known as:
(A) Isomorphism
(B) Allotropy
(C) Amorphism
(D) Polymorphism

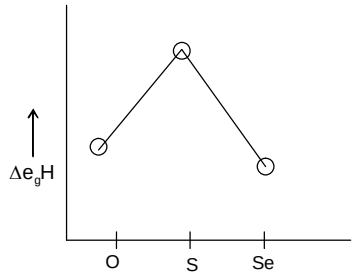
42. The INCORRECT statement amongst the following is:
(A) Boron has unusually high melting point due to strong crystalline lattice.
(B) In group 13, the compound with +1 O.S. are more ionic than with +3.
(C) Aluminium dissolves in dil. HCl to liberate H_2 whereas H_2 is not liberated when it reacts with NaOH.
(D) Aluminium chloride in acidified aqueous solution forms $\left[Al(H_2O)_6\right]^{+3}$ ion.

43. Pick out the correct statement:
(A) $\frac{r^+Cs^+}{r^+Cl^-}=0.536$.
(B) High temperature increases coordination number.
(C) Presence of excess Li^+ in LiCl develops violet colouration.
(D) Greater is the number of F-centres, greater is the intensity of colour developed.

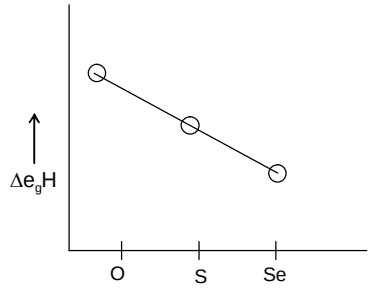
44. The INCORRECT order is:
(A) Covalent character: $PbCl_2 > CaCl_2 > SrCl_2 > BaCl_2$
(B) Thermal stability: $PbF_4 > PbCl_4 > PbBr_4 > PbI_4$
(C) Melting point: KF > KCl > KBr > KI
(D) Boiling point: $CHCl_3 > CH_3Cl > CCl_4$

45. The correct graph representing the electron gain enthalpy of chalcogens is: (Δe_gH is released energy)

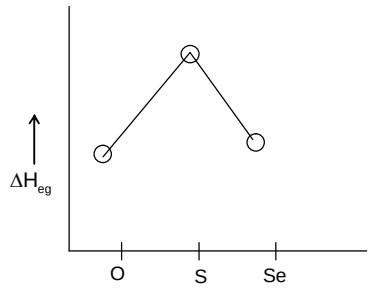
(A)



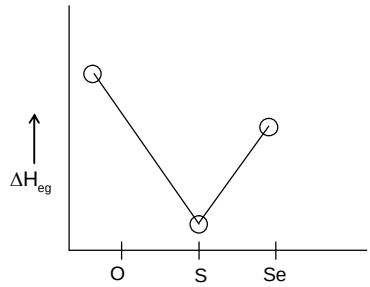
(B)



(C)



(D)



SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. The radial distribution curve of the orbital with double dumbshell shape in the fourth principal shell consists of 'n' nodes, n is:

47. After balancing the given reaction:

$$Ca_5(PO_4)_3F + SiO_2 + C \longrightarrow aP_4 + bCaF_2 + cCaSiO_3 + dCO$$

The value of [d – (a + b + c)] is:

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. Equal volumes of 0.02 M $AgNO_3$ and 0.02 M HCN were mixed. If $[Ag^+]$ at equilibrium is $x \times 10^{-5}$, calculate x. Take $K_a(HCN) = 9 \times 10^{-10}$ and $K_{sp}(AgCN) = 4 \times 10^{-16}$.

49. The ratio of distances of nearest neighbours in BCC lattice by next to next nearest in FCC lattice is:

50. The concentration of oxalic acid is 'X' mol L^{-1}. 40 ml of this solution react with 16 mL of 0.05 M acidified $KMnO_4$. What is the pH of 'X' M solution? Assume oxalic acid dissociates completely: (log2 = 0.3010).

Mathematics

PART – III

SECTION – A

(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. Let $\lambda = \int_0^1 \frac{dx}{1+x^3}$, $\beta = \lim_{n\to\infty}\left[\frac{\prod_{r=1}^{n}(n^3+r^3)}{n^{3n}}\right]^{\frac{1}{n}}$, then $\log \beta$ is equal to

(A) $\log 2 - 1 + \lambda$
(B) $\log 2 - 3 + 3\lambda$
(C) $2\log 2 - \lambda$
(D) $\log 4 - 3 + 3\lambda$

52. $\int \frac{x\cos\alpha + 1}{(x^2 + 2x\cos\alpha + 1)^{\frac{3}{2}}} dx$ is equal to

(A) $\frac{x}{\sqrt{x^2 + 2x\cos\alpha + 1}} + c$
(B) $\frac{-x\cos\alpha}{\sqrt{x^2 + 2x\cos\alpha + 1}} + c$
(C) $\frac{2x}{\sqrt{x^2 + 2x\cos\alpha + 1}} + c$
(D) $\frac{2x\cos\alpha}{\sqrt{x^2 + 2x\cos\alpha + 1}} + c$

53. $f(x) = \frac{[x]+1}{\{x\}+1}$ for $f:\left[0, \frac{5}{2}\right) \to \left(\frac{1}{2}, 3\right]$ (where [.] denotes the greatest integer function and {.} represents fractional part of x), then which of the following is true?

(A) f(x) is many-one discontinuous function
(B) $\min\left(\lim_{x\to 1^-} f(x), \lim_{x\to 1^+} f(x)\right) = f(1)$
(C) f(x) is surjective differentiable function every where
(D) f(x) is injective function

54. Let f(x) be a decreasing function defined on (0, ∞). If $f(2a^2 + a + 1) < f(3a^2 - 4a + 1)$, then the range of a is

(A) $a > 1$ or $a < \frac{1}{3}$
(B) $a \in (0, 5)$
(C) $a > \frac{1}{3}$
(D) $a \in \left(0, \frac{1}{3}\right) \cup (1, 5)$

55. Let f: R $\to$ R be a function such that f(0) = 1 and for x, y $\in$ R, f(xy + 1) = f(x)·f(y) – f(y) – x + 2 holds (where [.] denotes the greatest integer function). Then which is the correct option?

(A) $\lim\limits_{x\to 1^+}[f(x)] = 1$

(B) $\lim\limits_{x\to 2^+}[f(x)] = 2$

(C) $\lim\limits_{x\to 0^-}[f(x)] = 0$

(D) $\lim\limits_{x\to 0^-}[f(x)] = 1$

56. Tangent are drawn to the curve y = cos x from the origin. Then the point of contact lies on the curve $x^2 - y^2 = ax^2y^2$. This value of 'a' is equal to

(A) –1

(B) $\frac{1}{2}$

(C) 1

(D) 2

57. If the curve $y = x^2 + ax + b$ and $y = cx - x^2$ touch each other at (1, 0). Then the area bounded by the curve $y = x^2 + ax + b$ with x-axis is

(A) $\frac{2}{3}$

(B) $\frac{5}{6}$

(C) $\frac{1}{3}$

(D) $\frac{1}{6}$

58. Number of solutions of $\int_0^x \frac{dt}{\sqrt{4 - t^2 - t^3}} = x$ in (0, 1) is/are

(A) 1

(B) 2

(C) 3

(D) none of these

59. It is given that there are two sets of real numbers $A = \{a_1, a_2, ..., a_{100}\}$ and $B = \{b_1, b_2, ..., b_{50}\}$. Total number of onto functions 'f' from A to B such that $f(a_1) \le f(a_2) \le f(a_3) \le ..... \le f(a_{100})$ is/are

(A) $^{100}C_{50}$

(B) $^{99}C_{50}$

(C) $^{100}C_{49}$

(D) $^{99}C_{51}$

60. $\lim\limits_{x\to 0^+}\left(\left(x^{x^x}\right) - x^x\right)$ is equal to

(A) 0

(B) 1

(C) –1

(D) does not exist

61. If for all x, y the function f is defined by f(x) + f(y) + f(x)·f(y) = 1 and f(x) > 0, f(x) is differential everywhere, then
(A) $f'(0) > f'(1)$
(B) $f'(x) = 0$ for all x
(C) $f'(0) < f'(1)$
(D) none of these

62. If $f(x) = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!}(\log a)^n$, then at x = 0, f(x)
(A) has no limit
(B) is discontinuous
(C) is continuous but not differentiable
(D) is differentiable

63. $\lim_{n\to\infty} \cos\left(\pi\sqrt{n^2+n}\right)$, $n \in Z$ is equal to
(A) 0
(B) 1
(C) –1
(D) does not exist

64. n/m means that n is a factor of m, then the relation ' / ' is
(A) reflexive and symmetric
(B) transitive and reflexive
(C) transitive and symmetric
(D) equivalence

65. If f: $R \to R$ is a continuous function satisfying f(0) = 1 and $f(3x) - f(x) = x \ \forall\ x \in R$ and $\lim_{n\to\infty} f(x) - f\left(\frac{x}{3^n}\right) = P(x)$, then which of the following is correct?
(A) $P(\pi) = \pi$
(B) $P(e) = \frac{3e}{2}$
(C) P(5) = 10
(D) P(11) = 5.5

66. The value of $\int_0^{\infty} \frac{dx}{(1+x^3)(1+x^2)}$ is equal to
(A) $\frac{\pi}{3}$
(B) $\frac{\pi}{2}$
(C) $\frac{\pi}{6}$
(D) $\frac{\pi}{4}$

67. Let f(x) be a real valued differentiable function defined for all $x \geq 1$ such that $f'(x) = \frac{1}{x^2 + f(x)^4}$ and f(1) = 1, then

(A) $\lim\limits_{x\to\infty} f(x) < 1+\frac{\pi}{4}$

(B) $\lim\limits_{\lambda\to\infty} f(x) > 1+\frac{\pi}{4}$

(C) $\lim\limits_{x\to 0} f(x) > 1-\frac{\pi}{4}$

(D) we can't say

68. If α and β the roots of the equation $x^2 + t^2x - 2t = 0$, $I(t) = \int_{-1}^{2}\left[\left(x+\frac{1}{\alpha^2}\right)\left(x+\frac{1}{\beta^2}\right)+\frac{1}{\alpha\beta}\right]dx$, then

(A) $I(t) = \frac{3t^2}{8}+\frac{3}{4t^2}+3$

(B) $I(t) = \frac{3t^2}{8}-\frac{3}{4t^2}+3$

(C) $I(t) = -\frac{3t^2}{8}+\frac{3}{4t^2}-3$

(D) $I(t) = -\frac{3t^2}{8}+\frac{3}{4t^2}+3$

69. Let $\lim\limits_{x\to 0^+} f(x) = 24$ and $\lim\limits_{x\to 0^-} f(x) = 3$. If $\lim\limits_{x\to 0} f(2x^2 - x^3) = \lambda \lim\limits_{x\to 0} f(x^6 - x^2)$, then λ is equal to

(A) 27
(B) 21
(C) 12
(D) none of these

70. Let f: $(0, \infty) \to (0, \infty)$ be a differentiable surjective function and F(x) is the anti-derivative of f(x) such that $2(F(x) - f(x)) = f^2(x)$ for $\forall\ x \in R^+$, then

(A) $\lim\limits_{x\to\infty} \frac{f(x)}{x} = 2$

(B) $\lim\limits_{x\to\infty} \frac{f(x)}{x} = 1$

(C) f(x) is strictly decreasing function

(D) can't say

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. Let $M_n = \{0.a_1a_2 \ldots a_n$, such that $a_i = 0, 1$, for $1 \le i \le n-1$, $a_n = 1\}$, be a set of decimal fractions, T_n and S_n be the number and sum of the elements of M_n respectively, then $\lim\limits_{n\to\infty} \frac{54S_n}{T_n}$ is equal to

72. Let area bounded by the parabola $y^2 + 4y = 4x$ and $y = mx + m - 2$ is 9 sq. units, then value of $|9m|$ is equal to

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

73. The number of non-zero integral values of 'a' for which the function $f(x) = x^4 + ax^3 + \frac{27x^2}{2} + 1$ is concave upward for $\forall\ x \in R$

74. Let f(x) = ax + cos 2x + sin x + cos x is defined for $\forall\ x \in R$ and $a \in R$, f(x) is strictly increasing function the range of a is $\left[\frac{\alpha}{2}, \infty\right)$, then α is

75. Let $\int_0^1 (2x^2 - 3f(x))f(x)dx = \frac{1}{15}$ and $\int_0^1 f(x)dx = \frac{1}{k}$, then k is equal to